LLM ENGINEERING

Semantic Search Visual Guide

Finding meaning, not just keywords: dense vs sparse, hybrid, rerankers.

ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/5

Dense vs Sparse Retrieval

Meaning vectors vs keyword matching.

Dense

finds paraphrases,
misses exact terms.

embeddings + ANN:

Sparse

finds exact terms,
misses paraphrases.

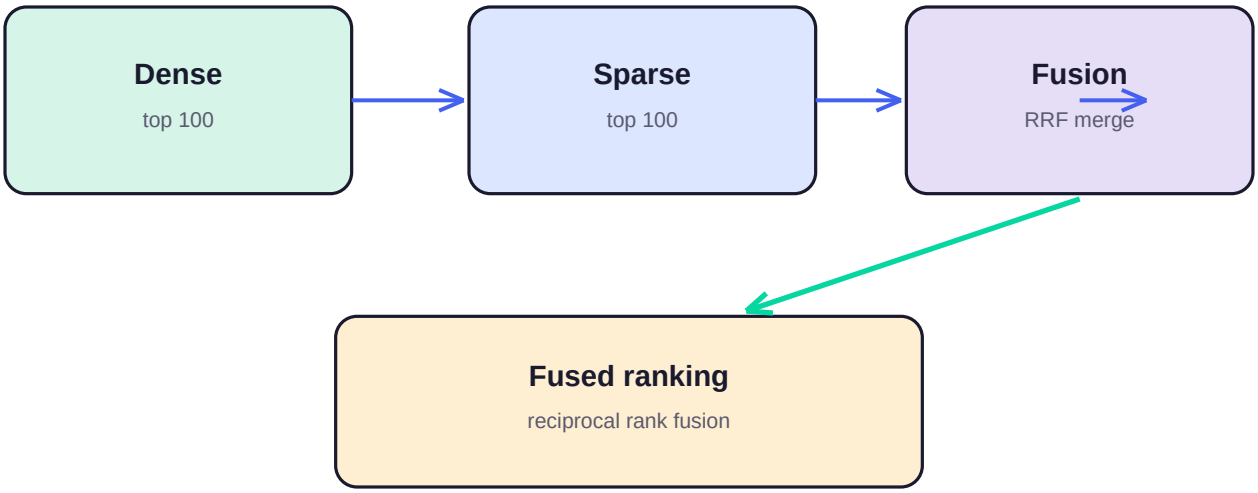
BM25 keywords:

They fail in opposite ways — which is exactly why combining them works so well.

STEP 2/5

Hybrid Search

Fuse both signals.



RRF: score = sum over lists of 1/(k + rank). No score normalization needed — ranks only.

Hybrid is the production default: dense for semantics, sparse for names, codes, and rare terms.

STEP 3/5

Query Understanding

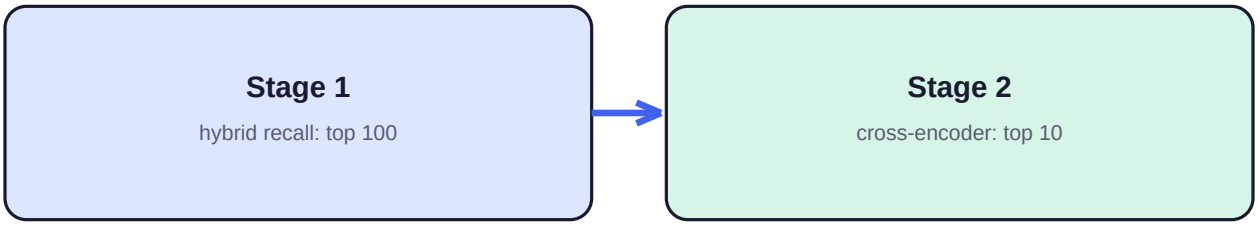
Fix the query before searching.

- Query expansion: add synonyms and related terms ('car' -> 'automobile, vehicle').
- HyDE: have an LLM draft a hypothetical answer, then search with its embedding.
- Query rewriting: turn a vague question into a precise search query with an LLM.
- Spell correction and entity recognition: 'iphon 15' should still find iPhone 15.
- Decompose multi-hop questions into sub-queries; search each, then synthesize.

STEP 4/5

Rerankers

A careful second look at the top candidates.



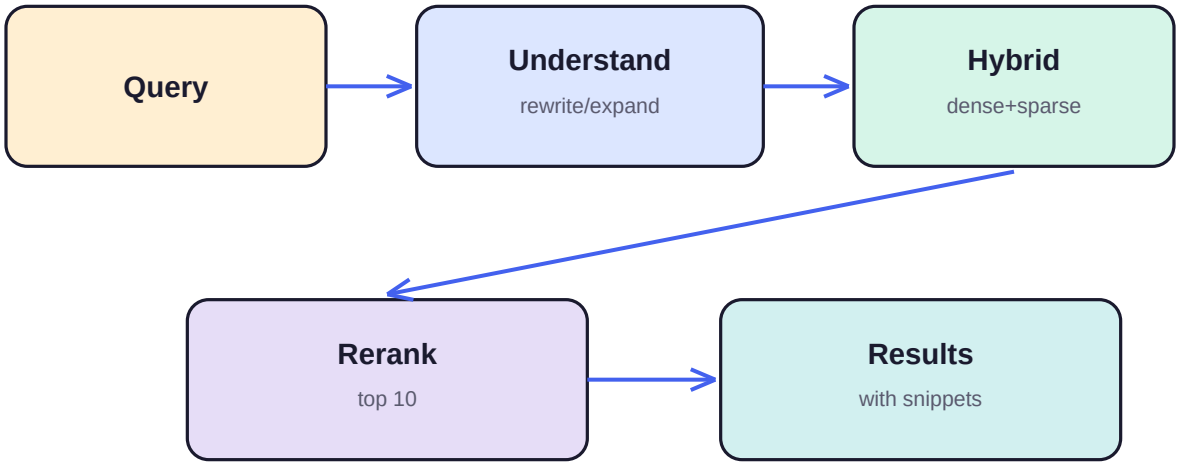
Cross-encoders score query+document jointly — far more accurate than bi-encoder similarity.

Reranking is the highest-ROI upgrade for most search systems: keep recall broad, let the reranker be picky.

STEP 5/5

End-to-End Pipeline

The full semantic search stack.



Instrument every stage: log queries, retrieved sets, and clicks. Search quality is measured, not assumed.

STEP ✓I✓

Key Takeaways

Semantic Search — Remember

- Dense finds paraphrases; sparse finds exact terms — they fail in opposite ways.
- Hybrid search with RRF fusion is the production default.
- Understand the query first: expand, rewrite, or HyDE before retrieving.
- Rerankers are the highest-ROI upgrade: broad recall, picky precision.
- Measure with labeled relevance judgments; log everything to improve.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev